Analyzing COVID-19 and Domestic Violence Incidence

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Agenda

01. **Policy Context and Problem**

02. **Methodology**

03. **Research Findings**

04. **Future Directions**

Check out our website!

https://helenw566.github.io/nyc_covid_dv/

How does incidence of DV change during pandemics?

Goal: Identify useful trends to inform future research about DV response measures

Findings from a natural experiment on the impact of covid-19 residential quarantines on domestic violence patterns in New Orleans

Auzeen Shariati [1,✉], Rob T Guerette [2]

Author information Article notes Copyright and License information

PMCID: PMC8885318 PMID: 35250165

Context and Problem Definition

- Literature review: COVID-19 Impacts in US Cities
- Framework development: Examining quarantine/restriction levels in New York City
- Existing DV reporting structures: gives a sense of DV reporting procedures before and after the periods we have chosen to examine

Methodology

Dataset: ENDGBV

End Domestic and Gender Based Violence: The Intersection of Domestic Violence, Race/Ethnicity and Sex (n = 48k)

- NYPD Incident Level Data collected between 2020-2021
- All incidents were domestic violence related, including
  - Felony Assaults
  - Rape
  - Domestic Incident Reports
- Largely categorical data, including time, victim and suspect demographics, geographic characteristics, etc.

COVID-19 REOPENING PHASES

1	2	3	4
June 8th, 2020	June 22nd, 2020	July 6th, 2020	July 20th, 2020

Methodology

01

Create an Amazon RDS and connect to it via Jupyter Notebook and R

02

Clean data, producing one non-imputed and one imputed table

03

Conduct preliminary exploratory data analysis

04

Run regression discontinuity analysis to assess impact of COVID-19 Reopening

Aggregate DV volume hides the real story. Offense severity matters.

Why stratify by offense type?

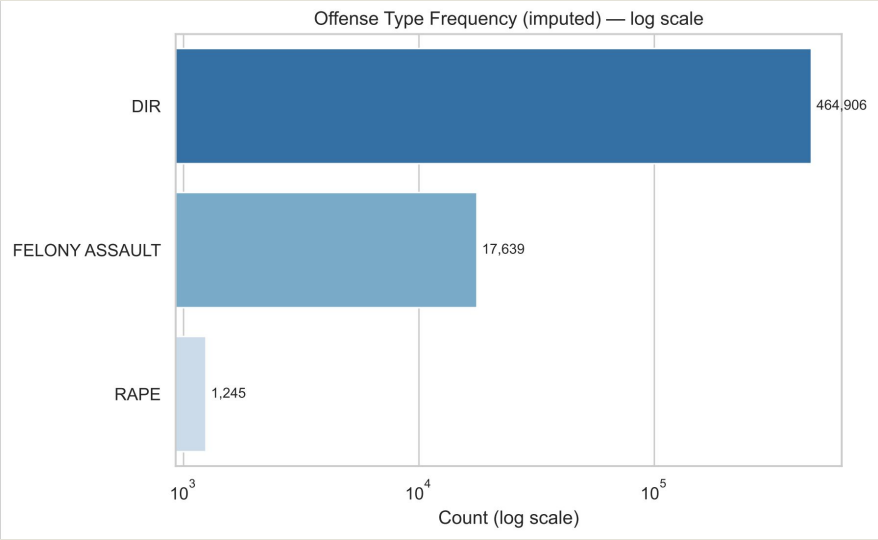

Figure 1. Offense type frequency (log scale, imputed data)

- DIRs outnumber Felony Assault by ~26x and Rape by ~370x.
- Pooling all three would let DIR volume drown out movement in the more severe offenses.
- We therefore run the RD separately for each offense type, further stratified by partner vs. non-partner offender.

Research Findings

Regression Discontinuity at Phase 1 Reopening

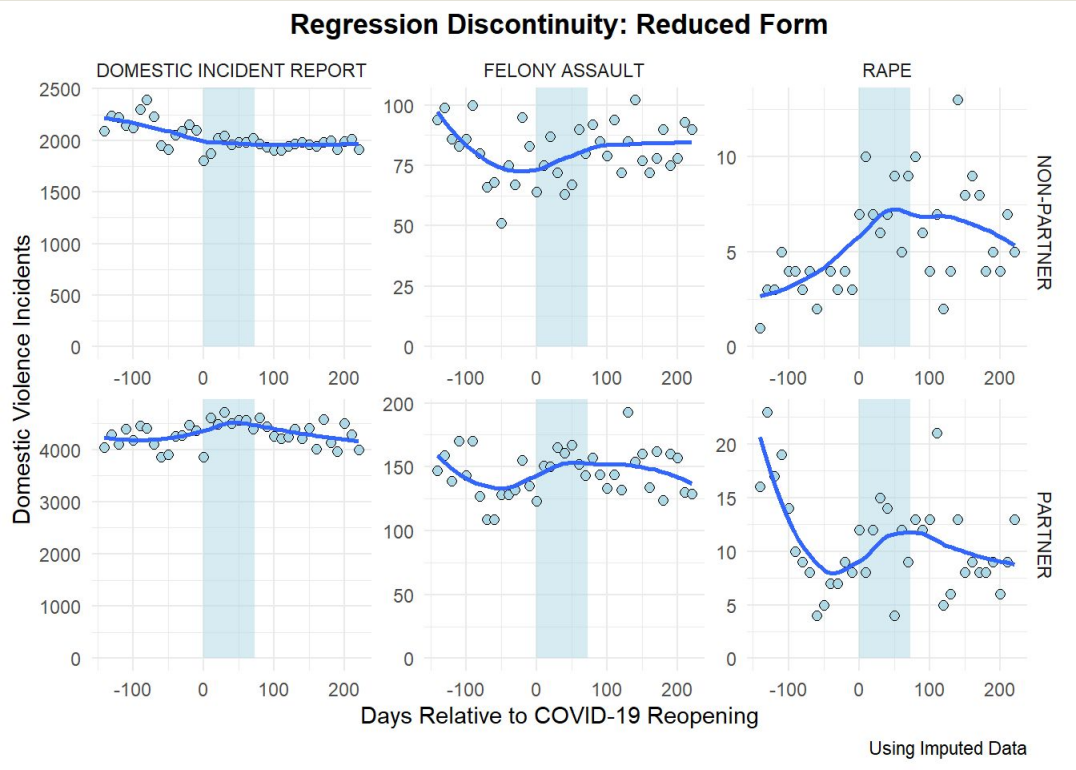

Figure 2. RD reduced-form fits by offense type (cols) × partner status (rows).

Partner Rape

β = 8.92 $p < 0.001$

Non-Partner Rape

β = 4.13 $p < 0.001$

Partner Felony Assault

β = 27.03 $p \approx 0.015$

Spike is at reopening, not at lockdown.

DIR estimates are weak and imputation-sensitive — aggregate volume masks these severity-specific shifts.

Why the Spike at Opening

During lockdown

Stay-at-home orders physically separated some perpetrators from victims (different households, constrained mobility).

Reported serious-offense rates stayed flat or declined across most panels.

At Phase 1 reopening (June 8)

Routine-activity patterns resume: contact, mobility, and exposure opportunities return

Reported partner-perpetrated rape and felony assault jump significantly at the cutoff.

Reopening, not lockdown, is the moment when routine-activity exposure restores and reported serious DV rises.

Conclusion

01

Reopening, Not Lockdown, Was the Inflection Point

Partner-perpetrated rape, non-partner rape, and partner felony assault all exhibit significant discontinuities at the June 8 cutoff — contradicting the assumption that lockdown was peak-risk.

02

Aggregate DV Volume Is a Misleading Signal

Domestic Incident Reports dominate the dataset but show weak, unstable RD estimates. The clearest increases appear in the most severe offenses — exactly where aggregate counts obscure rather than reveal.

03

Policy Transitions Require Targeted Surge Capacity

Outreach, shelter access, and hotline staffing should expand around policy transitions, not only during lockdown peaks. Severity-disaggregated monitoring should replace aggregate-volume dashboards.

Future Research Directions

Geographic Replication

Replicate this RD design in other cities with comparably well-defined reopening timelines to test whether the reopening-as-inflection-point pattern generalizes beyond NYC.

Extended Time Horizon

Extend analysis into later pandemic phases and the post-pandemic recovery period to determine whether post-reopening increases in partner violence persisted, decayed, or reverted.

Triangulate Outcome Sources

Cross-validate reported incidents against hotline call volumes, emergency room records, and shelter intake data to separate true incidence changes from reporting-behavior changes.

Subgroup & Spatial Analysis

Conduct finer-grained geographic and demographic subgroup analyses to identify which populations were most affected by policy transitions and strengthen the link to targeted policy response.

THANK YOU!